

EXPERIMENT 15

Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

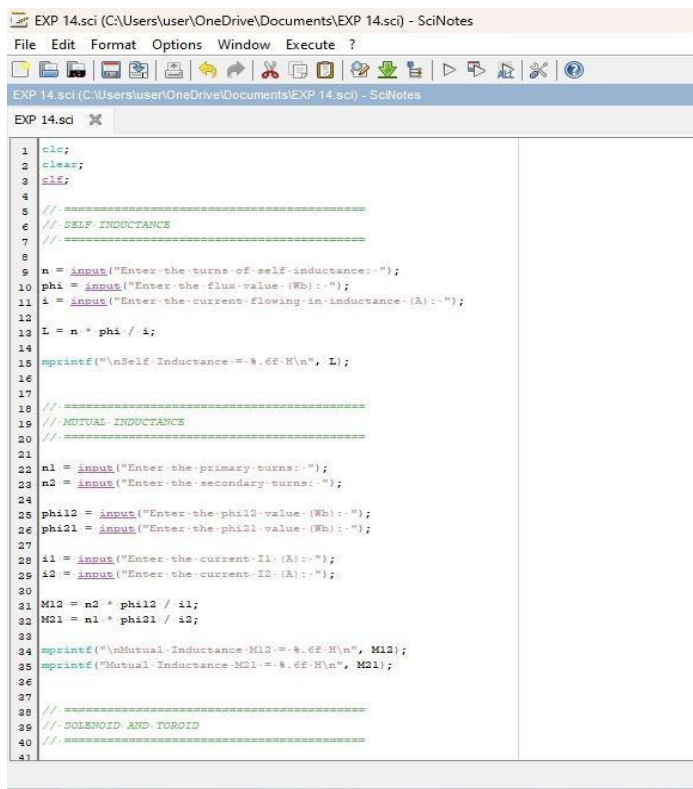
Aim:

To determine and plot Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

Software required:

SCILAB version 6.0.1.

Program:



```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1  clc;
2  clear;
3  clf;
4
5  // =====
6  // SELF INDUCTANCE
7  // =====
8
9  n = input("Enter the turns of self inductance:-");
10 phi = input("Enter the flux value (Wb):-");
11 i = input("Enter the current flowing in inductance (A):-");
12
13 L = n * phi / i;
14
15 mprintf("\nSelf Inductance:-%.6f H\n", L);
16
17
18 // =====
19 // MUTUAL INDUCTANCE
20 // =====
21
22 n1 = input("Enter the primary turns:-");
23 n2 = input("Enter the secondary turns:-");
24
25 phi12 = input("Enter the phi12 value (Wb):-");
26 phi21 = input("Enter the phi21 value (Wb):-");
27
28 i1 = input("Enter the current I1 (A):-");
29 i2 = input("Enter the current I2 (A):-");
30
31 M12 = n2 * phi12 / i1;
32 M21 = n1 * phi21 / i2;
33
34 mprintf("\nMutual Inductance M12:-%.6f H\n", M12);
35 mprintf("Mutual Inductance M21:-%.6f H\n", M21);
36
37
38 // =====
39 // SOLENOID AND TOROID
40 // =====
41
```

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
72 // =====
73 // PLOT ALL RESULTS
74 // =====
75 figure(1);
76
77 subplot(3,2,1);
78 plot(a, b, "-o");
79 xlabel("Time / - Variation");
80 ylabel("Self-Inductance - (H)");
81 title("Self-Inductance");
82 xgrid();
83
84 subplot(3,2,2);
85 plot(a, c, "-o");
86 xlabel("Time / - Variation");
87 ylabel("M12 - (H)");
88 title("Mutual-Inductance-M12");
89 xgrid();
90
91 subplot(3,2,3);
92 plot(a, d, "-o");
93 xlabel("Time / - Variation");
94 ylabel("M21 - (H)");
95 title("Mutual-Inductance-M21");
96 xgrid();
97
98 subplot(3,2,4);
99 plot(a, e, "-o");
100 xlabel("Time / - Variation");
101 ylabel("Solenoid");
102 title("Solenoid");
103 xgrid();
104
105 subplot(3,2,5);
106 plot(a, f, "-o");
107 xlabel("Time / - Variation");
108 ylabel("Toroid");
109 title("Toroid");
110 xgrid();
111
112
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Name

- Documents
- My Music
- My Pictures
- My Videos
- ex 6.sci
- EXP 14.sci

Enter the turns of self inductance: 55

Enter the flux value (Wb): 3×10^{-3}

Enter the current flowing in inductance (A): 5

Self Inductance = 297.000000 H

Enter the primary turns: 300

Enter the secondary turns: 400

Enter the ϕ_{112} value (Wb): 2×10^{-5}

Enter the ϕ_{121} value (Wb): 3×10^{-5}

Enter the current I1 (A): 5

Enter the current I2 (A): 6

Mutual Inductance M12 = 1200.000000 H

Mutual Inductance M21 = 1250.000000 H

Enter the length of solenoid (m): 5

Enter the radius of toroid (m): 5

Solenoid value = 1.256637×10^{-6}

Toroid value = 2.000000×10^{-7}

Variable Browser

Name	Value	Type	Visibility	Memory
L	297	Double	local	216 B
LENGTH	5	Double	local	216 B
M12	1.2×10^3	Double	local	216 B
M21	1.25×10^3	Double	local	216 B
Radius	5	Double	local	216 B
a	1x50	Double	local	608 B
ans	1x1 Graphic...	local	local	216 B
b	1x50	Double	local	608 B
c	1x50	Double	local	608 B

Command History

```
-- 30/08/2026 11:26:37 -- //
-- 55
-- 3*10^-3
-- 5
-- 300
-- 400
-- 2*10^-5
-- 3*10^-5
-- 5
-- 6
-- 5
```

News feed

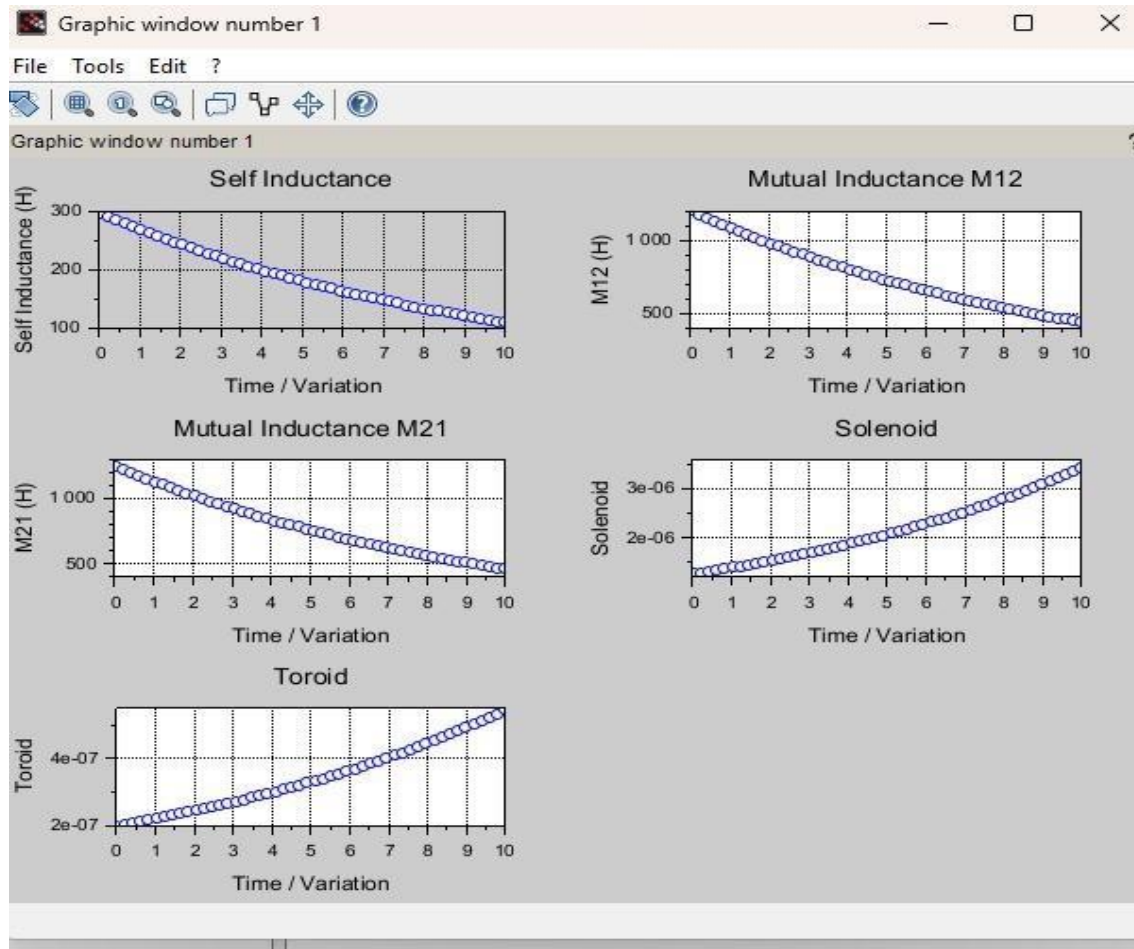
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- Publish toolboxes on [ATOMS](#).
- Post questions on [Discourse](#).

File/directory filter: Aa * * * *

11:41 AM 8/30/2026



Theoretical calculations:

Ex No 15

Self induction (m) = 2.50

$$\phi = 5 \times 10^{-4}$$

current in inductor (2) = 3 A

$$n_1 = 500$$

$$n_2 = 600$$

$$\phi_{12} = 2 \times 10^{-3}$$

$$\phi_{21} = 3 \times 10^{-3}$$

$$I_1 = 5 \text{ A}$$

$$I_2 = 6 \text{ A}$$

$$\text{length} = 3 \text{ cm}$$

$$\text{radius} = 3 \text{ mm}$$

i) Self inductance

$$L = \frac{N \phi}{I}$$

$$L = \frac{0.50 \times (5 \times 10^{-4})}{8}$$

$$= 25 \times 10^{-5} \text{ H}$$

$$= 3.125 \times 10^{-5} \text{ H}$$

2.

$\text{width} = 5 \text{ cm}$

$\text{height} = 3 \text{ mm}$

$\text{width} = 3 \text{ mm}$

2. Mutual inductance

$$M_{12} = \frac{N_2 \phi_{12}}{I_1}$$
$$= \frac{600 (2 \times 10^{-3})}{5}$$
$$= 0.24 \text{ H}$$

3. Mutual inductance

$$M_{21} = \frac{N_1 \phi_{21}}{I_2}$$

$$= \frac{500 \times (3 \times 10^{-3})}{1}$$

$$= 0.25 \text{ H}$$

O/P
✓

Result:

Thus determination and plotting of Inductance ,Self-inductance, Mutual inductance of current carrying coil, solenoid and toroid with different (a) length and (b) circumferential area are performed and verified successfully.